

Course Outline

FAST School of Management

National University of Computer & Emerging Sciences Lahore Campus



Course: Fundamentals of Programming for Business			
Course code:	CL1002	Year/Semester:	Fall 2025
Program:	BSBA	Units/Cr Hrs:	1 Cr Hr
Department:	FAST School of Management	Instructor:	Muhammad Azam
Course Type:	Core Course	Email:	muhammad.azam1@lhr.nu.edu.pk
Pre-Requisite(s):	None	Phone:	
Prepared By:		Consultation Hours:	
Approved By:		Approval Date:	

COURSE DESCRIPTION

In this course, students will gain understanding of fundamentals of computer programming. Students will acquire basic programming skills such as computational thinking, problem solving, basic data types, loops, functions etc. Through practical programming activities, students will gain an appreciation of the nature of computer programming. The focus is to develop the foundation and an aptitude for programming at advanced levels. Since, python is a general-purpose, versatile, and powerful programming language. It's a great first language because it's concise and easy to read. Whatever you want to do, Python can do it from web development to machine learning to data science.

COURSE OBJECTIVES

Upon successful completion of the course students should be able to:	
1	Comprehend the notion of computational thinking and understand the development of algorithms
2	Develop problem solving and logic building skills in students.
3	To introduce the basic concepts of programming in Python strengthening the foundation including basic data types, expressions, iterations, functions and arrays.
4	Provide hands-on experience of implementing basic programming concepts in Python programming language.

COURSE LEARNING OUTCOMES

LO#	Learning Outcome Statement	Program Objective	Bloom's Taxonomy Level
LO1	Understand basic problem-solving steps and logic constructs through computational thinking	PO1	Understand
LO2	Design and implement algorithms to solve real world business problems and should be able to translate a problem statement into pseudo-code/Python code	PO1, PO3, PO5,	Analyze, Evaluate
LO3	Apply basic programming concepts	PO1, PO5, PO10.4	Apply
LO4	Understand the GUI Programming and the real world applications of programming	PO1, PO5, PO10.4	Understand, Apply

LEARNING OUTCOME ASSESMENT STRATEGY

Assessment Tool	LO1	LO2	LO3	LO4
Assignments / Projects	25	25	25	50
Quizzes	25	0	0	0
Mid Term	50	25	25	0
Final Exam	0	50	50	50

* Percentages are meant to be viewed vertically.

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COURSE

CONTENTS

LO/Week	Contents
LO 1 Week 1	Introduction & Course Outline Discussions Introduction to Programming and Computational Thinking Four Pillars of Computational Thinking Concept of Algorithms
LO 2 & 3 Week 2	Input, Processing & Output Pseudo code Concept of a program Variables, Operators and Output Class Practice
LO 2 & 3 Week 3	Decision Structures & Boolean Logic Simple and Nested if/else statements Logical Operators & Boolean Variables Class Practice
LO 2 & 3 Week 4-5	Repetition Structures The While and For Loop Nested Loops Input Validation loops Class Practice
LO 2 & 3 Week 6-7	Functions Defining and Designing Functions Local and Global Variables Passing Arguments Class Practice
LO 3 Week 8	Files and Exceptions Introduction to processing files and records Class Practice
LO 3 Week 9- 10	Lists and Tuples Processing Lists Slicing Built-in function List Comprehension Use of tuples Dictionaries & Sets Class Practice
LO2 & 3 Week 11	Numpy Pandas Exploratory Data Analysis Class Practice
LO 2& 3 Week 12-13	Matplotlib Seaborn Class Practice
LO4 Week 14	Project Presentations

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S. No	Material /Activity	Details
1.	Final Project	Explore the fundamental concepts of programming with practical application during the class lectures, assignments and the project.
2.	Guest Speaker Session	Guest speakers from the local industry are invited to give talks on Python and its applications to enlighten us with the practical applications in the industry.
3.	Local & International Case Studies	Students are given local and international case scenarios to link the concepts and their use.
4.	Soft Wares	Familiarity with practical use of Python and its core concepts

ADMINISTRATIVE INSTRUCTIONS

GRADING POLICY

Grading Mechanism	Weight
Assignments & Quizzes	15
Project	15
Class Participation/Attendance	05
Mid Term Exam	15
Final Exam	50
Grading Criteria	
ABSOLUTE Grading	☒ RELATIVE Grading

** Final Grading will be relative grading, using inter-quartile range/MCA method.

Class Policies

- Please note that any exception for one student is unfair to all other students, so don't expect any.
- Please turn off and store away cell phones, Ipads, laptops, and other electronic devices.
- Talking during lecture is not permitted. It is disrespectful and disruptive to other class members and the instructor.
- If you miss a class, it is your responsibility to determine what was covered including any administrative announcements.

Studying

The proper way of studying for this class is following ADA policy; a short description of the same is as follows

- Ahead of the class, it is expected that you have read the relevant chapters from the textbook;
- During the class you are expected to follow the lecture, take notes and ask questions; and
- After every class you would review your notes and solve the end of chapter exercises and read the textbook.

An extensive set of practice problems is placed on the Xeon Server, make sure you cover them all. We might set a tutorial session from time to time to answer your queries about the problems/practice sets.

Attendance Policy

As you can see, almost every other session is a different topic and as such missing any classes will result in huge setback as you will lose substantial course content. Besides, owing to the interactive nature of the course, attendance is crucial. Therefore, only under unavoidable circumstances, student should think of missing a class. See university policy on minimum attendance allowed for final exams.

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Scholastic Dishonesty

In the classroom and in all other academic activities, students are expected to uphold the highest standards of academic integrity. Any form of scholastic dishonesty is an affront to the pursuit of knowledge and jeopardizes the quality of the degree awarded to all graduates.

What exactly is plagiarism?

- Plagiarism is a form of cheating.
- Plagiarism is using someone else's ideas or words and saying they are your own.

If you use material from a text and do not acknowledge the source, you are committing plagiarism.

Specifically, these behaviors are often regarded as plagiarism:

- Copying directly from a text, acknowledging the source but pretending that you are paraphrasing.
- Paraphrasing or copying directly from a text without acknowledging the source.
- Copying from another student's assignment with or without the student's knowledge.

The following behaviors are regarded as misconduct:

- Submitting the same assignment in two different papers.
- Getting someone else to write an assignment for you.

You are also involved in misconduct if you:

- Let another student copy from your own work. ☒ Write an assignment for another student.

Students who violate the rules on scholastic dishonesty are subject to disciplinary penalties, including the possibility of failure in the course and/or dismissal from the University. *Since dishonesty harms the individual, all students, and the integrity of the University, policies on scholastic dishonesty will be strictly and actively enforced.*